6.1 Experiment 1: complete particle–mesh operator action

Experiment 1 asks whether the operator that the integrator actually applies, $X \mapsto \Gamma_h X$, reproduces the periodic friction operator Γ of the screened longitudinal kernel $K(\mathbf{r}) = e^{-\kappa r} r^{-1} P(\mathbf{r})$, $\kappa = 1$, with the error predicted by the choice of (ξ, r_c, k_c, h, p). We follow the hierarchy used for the Landau particle method: static configurations first, so that no time-discretization error enters; then each source of error on its own; then the complete action. The model is not changed for the test. The friction is fully periodic, the $k = 0$ Fourier mode is retained in both the reference and the particle–mesh operator, and there is no free-space truncation and no energy null space. The only null space is the translation space $\mathcal{T} = \mathrm{span}\{T_\alpha\}_{\alpha=1}^{3}$, $T_\alpha = \mathbf{1}_N \otimes \mathbf{e}_\alpha$.

Why small dense-reference systems. Every error in this experiment is measured against a dense $3N \times 3N$ matrix Γ_{ref} and its full spectrum. The reference therefore has to be assembled pair by pair over many periodic images, and it has to be diagonalized. This restricts the experiment to $N \leq 256$. These sizes are chosen to make the reference computable; they are not a regime of interest in their own right. The two original test boxes are case A ($N = 30$, $L = 6$) and case B ($N = 40$, $L = 10$). Here L is the edge length of the periodic cube $[0, L)^3$, not a number of boxes. Their number densities are $n = N/L^3 = 0.139$ and 0.040, and $\kappa L = 6$ and 10. Both are single draws of independent uniform positions (seeds 7 and 13), with no minimum distance imposed; the closest pairs are at 0.395 and 0.589. Case B was taken large enough that the smallest nonzero lattice wave number $2\pi/L$ lies below the wave number at which the longitudinal eigenvalue of $\hat{K}^L$ changes sign. No physical argument selected $L = 6$ for case A: it is a small box in which the dense reference is cheap. The parameter sets used later in the paper were accepted when they met their tolerance on these two boxes. Two boxes are not evidence that the same parameters hold for other configurations, and we use them only as the starting point.

Configurations. Table 1 lists all configurations, including the shortest periodic (minimum-image) pair distance $d_{\min}$ of each. Since $\kappa = 1$, the column L also gives κL. The main set has $N = 128$ at the density of case A, $n_A = 30/216$, so $L = 9.732$. It contains three families, each with three independent seeds:

- *uniform*: independent uniform positions with no minimum distance;
- *perturbed lattice*: a bcc lattice of 4^3 cells ($N = 128$; an fcc lattice of 4^3 cells for $N = 256$), every site displaced by an isotropic Gaussian with $\sigma = 0.1\, d_{\mathrm{nn}}$, plus a random global shift. Here $d_{\mathrm{nn}} = 2.107$ (bcc) or 2.168 (fcc) is the nearest-neighbour distance;
- *clustered*: $N/16$ cluster centres drawn uniformly, and 16 particles per cluster drawn from an isotropic Gaussian of width $\sigma_c = 0.5$ around their centre.

For the clustered family, a draw is rejected if it comes closer than 0.25 to a particle already placed. This avoids near-coincident pairs, whose $1/r$ kernel would make $\|\Gamma_{\mathrm{ref}}\|_2$ arbitrarily large. The realised $d_{\min}$ is 0.251–0.259 ($N = 128$ and 256), so the constraint is active. For the uniform family, which has no constraint, $d_{\min}$ is 0.142–0.411. One configuration per family at $N = 256$ (with $L = 12.26$ at the same density) is a smaller representative check. All random numbers come from seeded generators; the seeds and generator parameters are stored with the positions.

Two box-length sets are kept apart, because they change different things.

- In the *fixed-N* set, $N = 64$ and $L \in \{4, 6, 8, 10, 12, 16\}$. Changing L therefore also changes the density, from $n = 1$ to 0.0156.

Table 1: Static configurations of Experiment 1. L is the edge length of the periodic cube, $n = N/L^3$ the number density, $d_{\min}$ the smallest periodic pair distance (range over seeds), $\kappa = 1$. A and B are the original two dense-reference boxes.

set	generation	N	$L = \kappa L$	$n = N/L^3$	$d_{\min}$	seeds
A	uniform, seed 7	30	6	0.139	0.395	1
B	uniform, seed 13	40	10	0.040	0.589	1
U128	uniform i.i.d.	128	9.73	0.139	0.142–0.411	3
L128	bcc 4^3, $\sigma = 0.1\, d_{\mathrm{nn}}$	128	9.73	0.139	1.208–1.366	3
C128	clusters of 16, $\sigma_c = 0.5$, $d \geq 0.25$	128	9.73	0.139	0.251–0.255	3
U256	uniform i.i.d.	256	12.3	0.139	0.148	1
L256	fcc 4^3, $\sigma = 0.1\, d_{\mathrm{nn}}$	256	12.3	0.139	1.011	1
C256	clusters of 16, $\sigma_c = 0.5$, $d \geq 0.25$	256	12.3	0.139	0.259	1
box-N	uniform i.i.d.	64	4–16 (6 values)	0.016–1.00	0.090–0.723	3
box-n	uniform i.i.d.	30–240	6–12 (4 values)	0.139	0.055–0.924	3

- In the *fixed-density* set, $n = n_A$ and $L \in \{6, 8, 10, 12\}$, so $N = 30, 71, 139, 240$.

Each set uses three uniform seeds per L.

Reference and its convergence. Γ_{ref} is built from the unsplit kernel by a direct periodic lattice sum,

$$(\Gamma_{\mathrm{ref}} X)_i = \sum_{j \neq i} \sum_{\mathbf{m} \in \mathbb{Z}^3,\ |\mathbf{r}_{ij} + \mathbf{m}L| \leq R} K(\mathbf{r}_{ij} + \mathbf{m}L)\,(X_i - X_j).$$

It uses neither a minimum-image approximation nor any splitting, and because nothing is removed from the sum the $k = 0$ mode is included automatically. The image radius was increased through $R = 15, 20, \ldots, 45$. Over all 44 configurations, the largest change relative to $R = 45$ is 2.5×10^{-12} at $R = 30$, 2.0×10^{-14} at $R = 35$, and 4.3×10^{-16} at $R = 40$, measured in $\|\cdot\|_2 / \|\Gamma_{\mathrm{ref}}\|_2$. This change follows the tail $e^{-\kappa R}$ until it reaches round-off. We take $R = 45$ as the reference. The ladder shares its inner shells, so it bounds only the image truncation, which is below 10^{-15}. Round-off in the common sum is checked independently by the split construction below, which reproduces Γ_{ref} to 6.7×10^{-15}–2.2×10^{-13}. Both levels are far below the smallest approximation error measured against the reference in this section, 2.3×10^{-10}. The reference is symmetric to 1.0×10^{-16} and annihilates the translations to 1.9×10^{-16}, both relative.

Split identity and coefficient accuracy. On nonzero separations, θ_S is evaluated in closed form and θ_L from its own s-integral (not as $e^{-\kappa r}/r - \theta_S$). The identity $\theta_S + \theta_L = e^{-\kappa r}/r$ (draft Eqs. (7)–(8)) then holds to 2.5×10^{-15} relative error. This was checked for $\xi \in \{0.5, 0.7, 1, 1.4, 2\}$, on a radial grid $10^{-3} \leq r \leq 20$, and on every pair image with $r \leq 12$ of the main and legacy configurations.

At operator level we combine two independent constructions:

- Γ_S from the real-space lattice sum of θ_S over $|\mathbf{r} + \mathbf{m}L| \leq 45$;
- Γ_L^{F} from the untruncated (mesh-free) lattice Fourier sum $V^{-1} \sum_{|\mathbf{k}| \leq k_c} \hat{K}^L(\mathbf{k}) \cos(\mathbf{k} \cdot \mathbf{r})$, with $k_c = 2\xi \cdot 6.5$ ($e^{-(k_c/2\xi)^2} \approx 2 \times 10^{-19}$), including $\mathbf{k} = 0$.

Their sum reproduces Γ_{ref} to between 6.7×10^{-15} and 2.2×10^{-13} (Fig. 1a). The largest values occur for the perturbed lattices, whose $\|\Gamma_{\mathrm{ref}}\|_2$ is smallest. The same level is the plateau of the

Fourier-cutoff sweep below, and it grows with the number of retained modes. It therefore belongs to the float64 comparison of a Fourier lattice sum with a real-space lattice sum, not to the split identity, which holds pointwise to 2.5×10^{-15}. A recorded check (case A, a perturbed lattice and a uniform configuration at $\xi = 0.7$ and 1) narrows this down. The floor (2.3×10^{-14}–1.5×10^{-13}) agrees to within 0.2% across three variants: float64 accumulation, 80-bit accumulation of the same inputs, and coefficients from the refined quadrature rules. The float64 and 80-bit sums differ by at most 7×10^{-16}. The floor is therefore set neither by the accumulation nor by the coefficient quadrature. Its remaining sources, the rounding of positions and wave vectors in the phases $\mathbf{k} \cdot \mathbf{r}$ and the real-space sum Γ_L it is compared with, were not separated. The coefficients themselves are accurate. With the Gauss–Legendre rules for $A(k)$, $B(k)$ and θ_L refined (twice the panels, 30 nodes per panel), these change by at most 1.6×10^{-15}, relative to $\|\hat{K}^L(k)\|_2$ or to $e^{-\kappa r}/r$, for $\xi \le 2$, $k \le 14\xi$ and $10^{-3} \le r \le 20$. The quadrature of θ_S agrees with its closed form to 1.0×10^{-15}.

Real-space and Fourier cutoffs. Each truncation is measured with no other approximation present:

- $\|\Gamma_S^{r_c} - \Gamma_S\|_2$, where Γ_S is the real-space lattice sum of θ_S over $|\mathbf{r} + \mathbf{m}L| \le 45$;
- $\|\Gamma_L^{k_c} - \Gamma_L\|_2$, where $\Gamma_L^{k_c}$ is the mesh-free lattice Fourier sum over $|\mathbf{k}| \le k_c$, and Γ_L is the real-space lattice sum of θ_L over $|\mathbf{r} + \mathbf{m}L| \le 45$ ($\theta_L = e^{-\kappa r}/r - \theta_S$ for $\xi r \ge 1.5$, its s-integral otherwise).

Both are normalized by $\|\Gamma_{\mathrm{ref}}\|_2$ (Fig. 1b,c). We fit $\log E = a - cx^2$ with $x = \xi r_c$ or $x = k_c/(2\xi)$, using only $x \ge 1$ and $E \ge 10^{-12}$, so that any round-off plateau is excluded. Steps at which a larger cutoff admits no new pair image or lattice shell are left out of the fit (see below).

At $\xi = 0.7$ the fitted Gaussian rates are:

- real space: median $c = 0.99$, range 0.98–1.03 over the eleven main and legacy configurations;
- Fourier space: median $c = 0.97$, range 0.96–1.00.

At $\xi = 1$ the medians are again 0.99 (range 0.98–1.03) and 0.97 (0.96–1.00). The $N = 256$ configurations fall in the same ranges. Removing the continuum prefactor of the tails (draft Eq. (17), $E \propto x\, e^{-x^2}$) by dividing by x overcorrects, giving medians of 1.05 (real space) and 1.03 (Fourier). Neither form gives $c = 1$ within the fit window, so the algebraic prefactor of the operator-norm error is not that of the continuum tail. We do not interpret c as a measured universal exponent.

The real-space error is still decreasing at the largest cutoff, $\xi r_c = 6.5$, where it is at most 9×10^{-17}; only three of the 28 curves level off near 10^{-16}. The Fourier error reaches a plateau of 1.2×10^{-13} ($\xi = 0.7$) and 1.6×10^{-13} ($\xi = 1$) in every configuration: this is the lattice-sum floor discussed above. The finite particle set makes the curves staircases at small cutoffs:

- on the perturbed lattices, 13 real-space steps with $\xi r_c \le 1.25$ admit no new pair image;
- for case A, 3 Fourier steps with $k_c/(2\xi) \le 1$ admit no new lattice shell.

These points keep their equal errors and are flagged in the data, not smoothed. The seven of them that fall inside a fit window are excluded from the fits.

Mesh spacing, B-spline order and aliasing. With $\xi = 0.7$ and the production Fourier cutoff $k_c = 5.74$ ($s = 4.1$), we compare the particle–mesh long-range operator with the mesh-free lattice sum over the same modes. The difference $\Gamma_L^{\mathrm{mesh}} - \Gamma_L^{k_c}$ is then pure mesh error, with no truncation in it. For the eleven main and legacy configurations the scan covers $p = 3, \ldots, 8$ and nine FFT-friendly meshes from $\eta = k_c h/\pi \approx 0.91$–$0.99$ down to 0.23–0.25. Each mesh is evaluated at eight grid shifts: the unshifted grid and seven random global offsets in $[0, h)^3$ (Fig. 1d–f). The $N = 256$ checks use $p = 5, 7$, six meshes (η down to 0.23) and four shifts. The scan was repeated at a converged cutoff, $s = 5.5$ ($e^{-s^2} \approx 7 \times 10^{-14}$), for $p = 4, 6, 8$, seven meshes and eight shifts, on cases A and B and seed 1 of each $N = 128$ family. Adjacent meshes differ by 7–41% in h.

We fit the RMS over shifts on the four finest meshes with error at least 10^{-12}. The threshold excludes meshes on which the error approaches the $\sim 10^{-13}$ floor of the Fourier lattice sum it is compared with: the finest meshes for $p = 8$ at $s = 5.5$ (the error falls to 8.9×10^{-14} at $\eta = 0.25$), and at $s = 4.1$ the finest mesh for one configuration (uniform, seed 3) at $p = 8$. At $s = 4.1$ the fit windows therefore lie between $\eta = 0.23$ and 0.50, typically 0.25–0.40; at $s = 5.5$ they reach up to $\eta = 0.77$. The fitted slopes in h are, as medians with ranges over the eleven main and legacy configurations ($s = 4.1$) and over the five configurations of the $s = 5.5$ scan:

p	slope at $s = 4.1$, median (range)	slope at $s = 5.5$, median
3	3.09 (2.90–3.22)	–
4	4.07 (3.85–4.36)	4.10
5	5.24 (4.88–5.43)	–
6	6.19 (6.13–6.44)	6.25
7	7.46 (7.03–7.66)	–
8	8.50 (8.42–8.88)	8.87

For $p = 5, 7$ the three $N = 256$ checks give 5.12–5.39 and 7.44–7.58, inside these ranges. In the fitted windows the error is therefore $O(h^p)$ for every family. The median slopes exceed p for $p \geq 5$, by up to 1.0 for $p = 8$ (one $p = 5$ fit is 4.88). On the coarse meshes ($\eta = 0.55$–0.99) the decay is clearly faster than h^p: the median exponents there are 3.31, 4.53, 5.97, 7.30, 8.68 and 10.18 for $p = 3, \ldots, 8$, or 1.10–1.27 p. Both observations are consistent with the leading alias image (draft Eqs. (54), (58)). Its factor q^p, with $q = \eta_k/(2 - \eta_k)$ and $\eta_k = kh/\pi \leq \eta$ for a retained mode k, has local exponent between p and $2p/(2 - \eta)$ in h, and reaches p only as $\eta \to 0$. Slopes between adjacent meshes are too noisy to show this directly. The h^p law is thus observed for $\eta \approx 0.23$–0.50; its asymptotic constant has not been reached, and we report no such constant.

The phase sensitivity is moderate. Over all eight-shift scans, the largest error over shifts is at most 1.30 times the RMS and at most 1.92 times the smallest. Fitted on the same windows, the slope of the maximum differs from that of the RMS by -0.26 to $+0.20$. On the unshifted grid, the complete off-grid decomposition

$$K_h - K_F = V^{-1} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}_{ij}} \left[D_h + H_h(\alpha_i^* + \alpha_j + \alpha_i^* \alpha_j) \right]$$

reproduces the particle–mesh pair response to 3×10^{-15} of $\max |K_F|$. The literal spread–FFT–influence–gather response, computed one source particle at a time, agrees with the mode-space pair response to 1.3×10^{-15}. We checked this for $p = 4, 7$ on two meshes for case A, case B and seed 1 of each $N = 128$ family. In the decomposition:

- the linear cross terms $\alpha_i^* + \alpha_j$ between the principal term and the aliases carry 0.996–1.003 of the mesh error (spectral norm);

Table 2: Parameter sets of the production action test (accepted at $\xi = 0.7$ for the stated tolerance on cases A and B). M is the next $2, 3, 5$-smooth mesh size with $h \le \eta_N \pi / k_c$; the realised $\eta = k_c h / \pi$ is therefore at most η_N and depends on L.

set	ξ	s	$r_c = s/\xi$	$k_c = 2\xi s$	η_N, p	M (A / B / N=128 / N=256)
T5	0.7	3.494	4.991	4.891	0.8, 6	12 / 20 / 20 / 24
T7	0.7	4.100	5.857	5.740	0.7, 7	16 / 27 / 27 / 36
T9	0.7	4.628	6.611	6.479	0.6, 8	24 / 36 / 36 / 45

- the alias–alias term carries at most 0.9%;
- D_h vanishes to 5×10^{-9} of the error, as it should with ideal deconvolution.

An estimate that kept only the alias–alias term would therefore miss the mesh error almost completely.

The complete production action. We now apply the production code path itself: the real-space pair loop, B-spline spreading of the three components, FFT, multiplication by the influence function $\hat{K}^L/|\hat{W}|^2$ on $|\mathbf{k}| \le k_c$, inverse FFT, adjoint gather, and the degree correction. We use it in two ways:

- applied to the $3N$ unit vectors, it gives Γ_h column by column, without any hand-assembled matrix;
- applied to probe vectors, it gives the action directly.

Three parameter sets were tested (Table 2): the cheapest sets accepted at $\xi = 0.7$ for the tolerances 10^{-5}, 10^{-7} and 10^{-9} on cases A and B (T5, T7, T9). The literal columns agree with an independent mode-space assembly to 1.1×10^{-15}. That assembly, in turn, equals an independently built real-space cutoff sum $\Gamma_S^{r_c}$ plus the same mesh pair matrix Γ_L^{mesh} to 1.4×10^{-15}. The budget terms $\varepsilon_S = \|\Gamma_S^{r_c} - \Gamma_S\|_2$, $\varepsilon_F = \|\Gamma_L^{k_c} - \Gamma_L\|_2$ and $\varepsilon_{\text{mesh}} = \|\Gamma_L^{\text{mesh}} - \Gamma_L^{k_c}\|_2$, all normalized by $\|\Gamma_{\text{ref}}\|_2$, therefore telescope to the production error. All three sets keep $\mathbf{k} = 0$ and exclude no influence mode.

Figure 2a shows the relative spectral error $\|\Gamma_h - \Gamma_{\text{ref}}\|_2 / \|\Gamma_{\text{ref}}\|_2$. Cases A and B meet their tolerances, as they were selected to. The uniform and clustered configurations at $N = 128$ and 256 meet them with margin. The perturbed lattices do not:

- T5: 1.16–1.29×10^{-5};
- T7: 1.94–2.11×10^{-7} at $N = 128$ and 1.09×10^{-7} at $N = 256$;
- T9: 1.96–2.55×10^{-9}.

The excess is up to a factor 2.6.

Figure 2b shows the reason. At T7, the absolute error $\|\Gamma_h - \Gamma_{\text{ref}}\|_2$ lies in 5.7×10^{-8}–5.5×10^{-7} over all 44 configurations, a spread below a factor ten. It still depends systematically on the configuration: it is 3.4–5.2×10^{-7} for the clustered and 1.0–1.5×10^{-7} for the perturbed-lattice configurations, and it grows with density (see below). Meanwhile $\|\Gamma_{\text{ref}}\|_2$ ranges from 0.72 to 35, set by the closest pairs through $\theta(d_{\min}) \approx e^{-d_{\min}}/d_{\min}$. Perturbed lattices have no pair closer than 1.0 and the smallest norms (0.72–0.94). A tolerance stated relative to $\|\Gamma_{\text{ref}}\|_2$ and calibrated on

configurations that contain close pairs therefore does not transfer to configurations without them: the absolute error varies far less than the relative one, which inherits the configuration dependence of the norm.

For the main and legacy configurations at T7 the mesh term is the largest in the budget ($\leq 1.5 \times 10^{-7}$ relative), followed by the real-space and Fourier truncations ($\leq 1.0 \times 10^{-7}$ and $\leq 7 \times 10^{-8}$, largest for the lattices). The real-space term is comparable or larger for the $N = 256$ lattice and for the boxes with the smallest realised η (fixed-N $L = 4$, fixed-density $L = 8$), and at T5 it is the largest for the uniform and lattice configurations.

The probes are applied through the production matvec (Fig. 2c):

- sixteen mean-free Gaussian vectors;
- the lowest nonzero periodic wave number $2\pi/L$ along each axis, longitudinal and transverse, cosine and sine, projected onto $\mathcal{T}^{\perp}$;
- the three translations.

At T7 the largest relative action errors $\|(\Gamma_h - \Gamma_{\mathrm{ref}})X\|/\|\Gamma_{\mathrm{ref}}X\|$ on the main and legacy sets are 1.7×10^{-7} (Gaussian), 2.7×10^{-7} (longitudinal) and 4.2×10^{-7} (transverse), against a largest relative spectral error of 2.1×10^{-7}. A relative action error can exceed the configuration's own relative spectral error by at most the factor $\|\Gamma_{\mathrm{ref}}\|_2\|X\|/\|\Gamma_{\mathrm{ref}}X\|$. For the long-wavelength modes, which lie close to the null space of the graph Laplacian, this factor is large: $\|\Gamma_{\mathrm{ref}}X\|/(\|\Gamma_{\mathrm{ref}}\|_2\|X\|)$ is as small as 2.5×10^{-2} (longitudinal) and 1.1×10^{-2} (transverse) on the main and legacy sets. Every longitudinal and transverse probe exceeds its configuration's relative spectral error, by up to a factor 14 and 18. Only about half of the Gaussian probes do, by up to a factor 6. On the perturbed lattices, whose spectral error is largest, the Gaussian errors stay below it. The box-length sets reach smaller ratios (6.5×10^{-3} and 2.9×10^{-3}, for a configuration with a pair at 0.055 and $\|\Gamma_{\mathrm{ref}}\|_2 = 35$), with correspondingly larger factors. For the translations we report operator-normalized residuals instead of dividing by a zero reference action:

- the production $\Gamma_h T_\alpha$ is exactly zero in floating point, because the degree correction cancels the constant field term by term;
- $\|\Gamma_{\mathrm{ref}}T_\alpha\|/(\|\Gamma_{\mathrm{ref}}\|_2\|T_\alpha\|) \leq 1.9 \times 10^{-16}$.

The literal Γ_h is symmetric to 1.0×10^{-16} (relative Frobenius norm). For Experiment 3 we record the smallest eigenvalue on $\mathcal{T}^{\perp}$. It ranges over 8×10^{-4}–1.92 across the configurations, and is changed by Γ_h by at most 4.2×10^{-5} (T5), 1.6×10^{-6} (T7) and 2.6×10^{-8} (T9) relative. In every case $\|\Gamma_h - \Gamma_{\mathrm{ref}}\|_2 < \lambda_{\min}(\Gamma_{\mathrm{ref}}|_{\mathcal{T}^{\perp}})$, with margin at least $0.997\,\lambda_{\min}$.

Box length. At T7, the two box-length sets give different information (Fig. 2d).

All values below are medians over the three seeds.

At fixed $N = 64$ the density changes with L. The absolute error falls from 5.3×10^{-7} ($L = 4$, $n = 1$) to 1.0×10^{-7} ($L = 16$, $n = 0.016$). All three budget terms fall with the density:

- the absolute real-space term goes from 3.2×10^{-7} to 2.6×10^{-8} (sublinear in n);
- the Fourier term goes from 1.2×10^{-7} to 1.8×10^{-8};
- the mesh term goes from 2.9×10^{-7} to 8.8×10^{-8}; it is the largest term for $L \geq 6$, and it is also modulated by the realised η.

Rounding M to a $2, 3, 5$-smooth size gives $\eta = 0.61$ at $L = 4$ and 8 but 0.65–0.69 otherwise. This rounding causes the dip at $L = 8$.

At fixed density n_A the absolute error is 2.1×10^{-7} ($L = 6$), 1.1×10^{-7} ($L = 8$, $\eta = 0.61$), 2.5×10^{-7} ($L = 10$) and 2.5×10^{-7} ($L = 12$). The dip at $L = 8$ is again the mesh rounding. At nearly equal η (0.68–0.69), $L = 10$ and 12 lie about 20% above $L = 6$, beyond the seed spread (2.0–2.1×10^{-7} against 2.3–2.7×10^{-7}), mostly in the mesh term (1.9 against 2.3×10^{-7}). We did not isolate the cause of this residual L dependence.

A fixed-N comparison is therefore mainly a density test. Only the fixed-density comparison isolates L, and over $L = 6$–12 it shows changes of at most a factor 2.2, dominated by the realised η. The relative error additionally inherits the configuration-dependent $\|\Gamma_{\text{ref}}\|_2$, as above.

What the experiment supports. The original cases A and B show that the accepted parameter sets meet their relative tolerances on two uniform boxes. On their own they do not support a claim about other configurations. The supplementary experiment adds the following:

1. an independent reference built from the unsplit kernel, with $k = 0$, whose image truncation converges to below 10^{-15} and which an independent split construction reproduces to $\leq 2.2 \times 10^{-13}$;

2. the split identity and the coefficient quadrature at round-off over a range of ξ;

3. the Gaussian decay of both truncations and the $O(h^p)$ mesh law on $\eta \approx 0.23$–0.50, with consistent rates across three configuration families, two particle numbers and grid shifts, and with the linear alias cross terms identified as the mesh error;

4. a verification of the production spread–FFT–influence–gather action itself, including null-space and symmetry properties.

It also shows a limitation of the original acceptance: tolerances defined relative to $\|\Gamma_{\text{ref}}\|_2$ are configuration dependent and are exceeded by up to a factor 2.6 on perturbed lattices. The absolute error varies by less than a factor ten across all configurations, with a systematic dependence on clustering and density, far less than the relative error does. Parameter selection for production runs should therefore be calibrated on the configuration class with the smallest $\|\Gamma\|_2$, or on an absolute scale.

All results are for $\kappa = 1$, $N \leq 256$ and static configurations. The h^p rate is observed for $\eta \approx 0.23$–0.50; on coarser meshes the decay is faster (1.1–$1.3\,p$), and the asymptotic constant of the law was not reached.

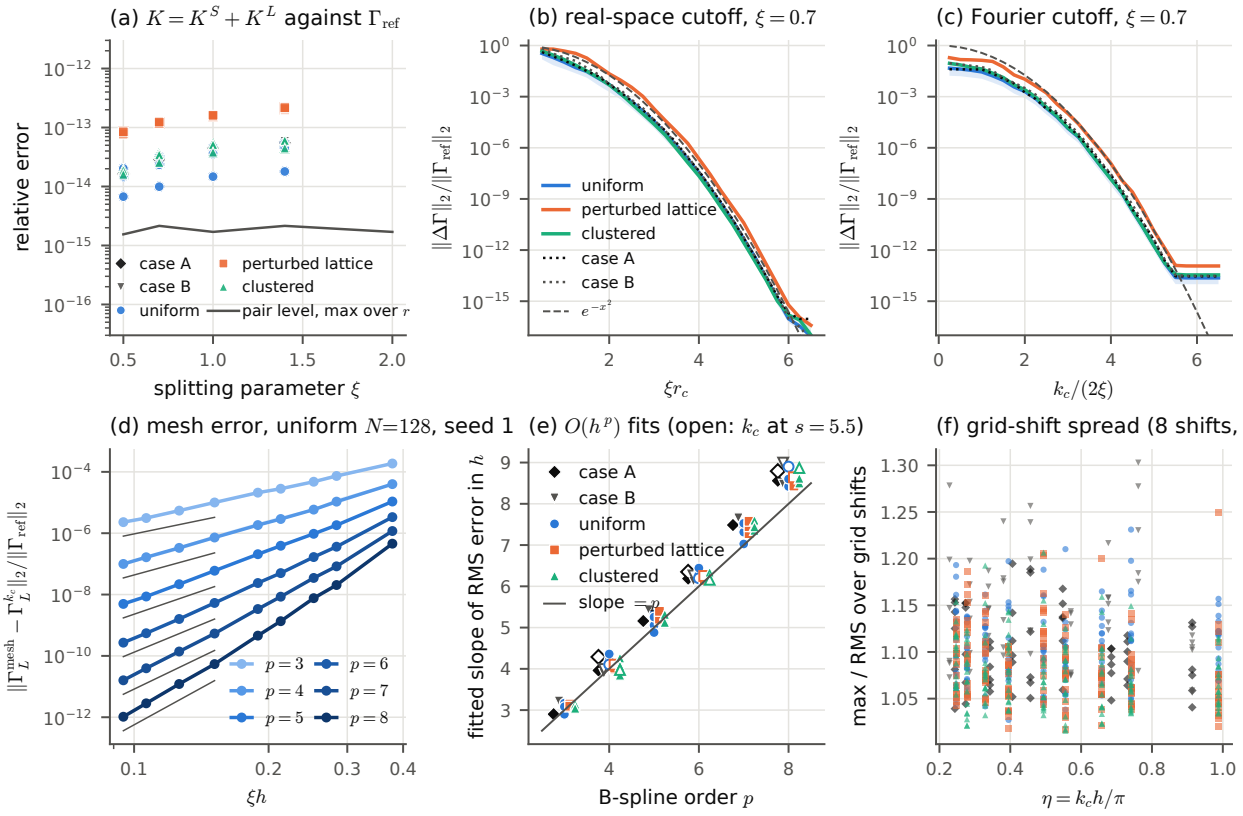


Figure 1: Error sources of Experiment 1, $\kappa = 1$. (a) Split identity: Γ_S (real-space lattice sum) plus Γ_L^{F} (mesh-free lattice Fourier sum, $k_c = 2\xi \cdot 6.5$) against Γ_{ref}. The line is the pair-level error $\max_r |\theta_S + \theta_L - e^{-\kappa r}/r|/(e^{-\kappa r}/r)$. (b) Real-space and (c) Fourier truncation at $\xi = 0.7$, with no other approximation present. The curves are the median over three seeds and the band is their range; dotted curves are cases A and B; the dashed curve is e^{-x^2}. (d) Mesh error against the mesh-free lattice sum over the same modes at the production $k_c = 5.74$, for uniform $N = 128$, seed 1. The curves are the RMS over eight grid shifts, the bands the range over shifts, and the thin segments have slope p. (e) Fitted slopes in h on the four finest meshes with error $\geq 10^{-12}$. Filled markers use the production k_c; open markers use k_c at $s = 5.5$ (five configurations). (f) Largest over RMS error across the eight grid shifts, for every mesh and p of the eight-shift scans ($N \leq 128$, production k_c).

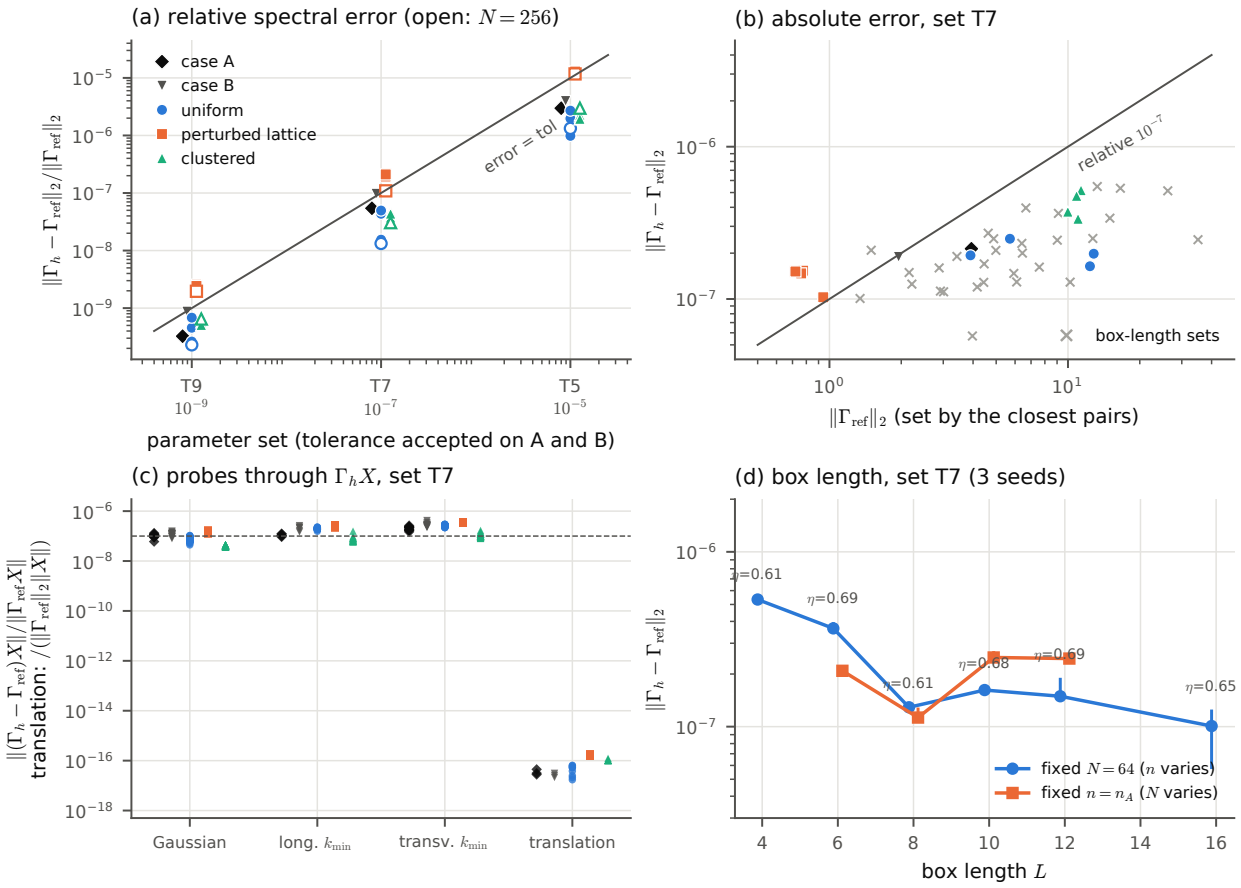


Figure 2: The production action $X \mapsto \Gamma_h X$ against the dense full-periodic reference. (a) Relative spectral error for the parameter sets accepted at 10^{-9}, 10^{-7} and 10^{-5} on cases A and B. Filled markers are $N \leq 128$ and open markers $N = 256$. (b) Absolute error against $\|\Gamma_{\mathrm{ref}}\|_2$ at T7; the diagonal is relative 10^{-7}; crosses are the box-length sets. (c) Probe errors through the production matvec at T7. Translations are given as operator-normalized residuals; the dashed line is 10^{-7}. (d) Absolute error against box length at T7, at fixed $N = 64$ (density varies) and at fixed density n_A (N varies). The realised η is annotated.